

USING LOOK, LISTEN, AND LEARN EMBEDDINGS FOR DETECTING ANOMALOUS SOUNDS IN MACHINE CONDITION MONITORING

Kevin Wilkinghoff

Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE
 Fraunhoferstraße 20, 53343 Wachtberg, Germany
 kevin.wilkinghoff@fkie.fraunhofer.de

ABSTRACT

The goal of anomalous sound detection is to unsupervisedly train a system to distinguish normal from anomalous sounds that substantially differ from the normal sounds used for training. In this paper, a system based on Look, Listen, and Learn embeddings, which participated in task 2 “Unsupervised Detection of Anomalous Sounds for Machine Condition Monitoring” of the DCASE challenge 2020 and is adapted from an open-set machine listening system, is presented. The experimental results show that the presented system significantly outperforms the baseline system of the challenge both in detecting outliers and in recognizing the correct machine type or exact machine id. Moreover, it is shown that an ensemble consisting of the presented system and the baseline system performs even better than both of its components.

Index Terms— anomalous sound detection, machine listening, deep audio embeddings, outlier detection

1. INTRODUCTION

Anomalous sound detection has many applications. Examples are detecting accidents in audio streams of road surveillance systems [1, 2], detecting screams or breaking glass as indicators of terror attacks in subway stations [3] or detecting mechanical failure in factories [4]. However, gathering anomalous data for training automatic systems is difficult because these events rarely occur and are very diverse. Thus, a system is trained unsupervisedly using normal data only and its task is to detect anomalous data that substantially differs from the training data. This task is known as outlier detection [5]. Among the models that are used for detecting anomalous sounds are one-class SVMs [6], convolutional neural networks as for example WaveNet [3] and many types of autoencoders [7] as autoencoders with a specific objective function [8, 9], autoencoders in combination with a bidirectional long short-term memory (BLSTM) [2] or denoising autoencoders with a BLSTM [10]. In [7], it has also been shown that enhancing sound quality by dereverberation and denoising before applying an ensemble of deep autoencoders is beneficial.

The goal of this paper is to use Look, Listen, and Learn (L^3 -Net) embeddings [11, 12] for detecting anomalous sounds when monitoring machine conditions. For this purpose, a recently proposed open-set machine listening system based on L^3 -net embeddings [13] is utilized and adapted accordingly. All experiments are conducted within task 2, titled “Unsupervised Detection of Anomalous Sounds for Machine Condition Monitoring”, of the DCASE challenge 2020 [14]. The dataset of the task is divided into a development set, an additional training set and an evaluation set. The development set consists of audio recordings from 4 different machines for each machine type and is divided into a training set with

around 1000 normal samples per machine and a test set with 100 to 200 normal and anomalous sounds. It is worth emphasizing that this test set is not allowed to be used for training the final system submitted to the challenge. This means that only normal samples are allowed to be used for training. The additional training set consists of audio recordings from 3 different machines for each machine type with around 1000 additional normal audio samples. These machines are different from the ones of the development set. The evaluation set consists of around 400 samples for each machine present in the additional training set and contains normal as well as anomalous samples. In total, the dataset contains six different machine types, namely “fan”, “pump”, “slide rail”, “valve” from MIMII [4] and “toy-car”, “toy-conveyor” from ToyADMOS [15]. Each audio file has a length of 10s with a sampling rate of 16kHz.

The contributions of this paper are the following. First and foremost, an anomalous sound detection system based on look, listen, and learn embeddings is presented, which is adapted from an open-set machine listening system [13]. Second, the system is compared to the baseline system of task 2 of the DCASE 2020 challenge. It is shown that the system based on L^3 -Net embeddings performs significantly better when detecting anomalous sounds and when predicting the machine type or exact machine id of recorded sounds. As a third contribution, an ensemble of both system is proposed, which performs better than both subsystems.

2. SYSTEM DESCRIPTION

2.1. Baseline system

The baseline system consists of multiple autoencoders each belonging to another machine type. After training, the reconstruction loss of the autoencoder belonging to the correct machine type is utilized as an anomaly score: a low loss corresponds to a normal machine sound and a high loss to an anomalous one. The input of the autoencoders are stacked frames of log-Mel spectrograms. More concretely, an audio file is converted into a log-Mel spectrograms using a frame size of 64ms, a hop size of 50% and 128 Mel bins. Then, for each frame its P preceding and P following frames are concatenated into a single vector. In all experiments, P is set to 2 and thus the input dimension is $128 \cdot (2P + 1) = 640$. All autoencoders consist of 4 encoding layers with a dimension of 128, a code layer of dimension 8, 4 decoding layers with a dimension of 128 and an output layer of dimension 640. In each layer but the output layer, a rectified linear unit (ReLU) is used as a nonlinearity and batch normalization [16] is applied. Every network is trained for 100 epochs with a batch size of 512 using Adam [17] and is implemented via Keras [18] and Tensorflow [19].

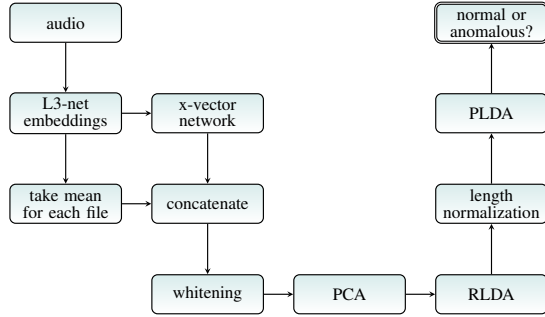


Figure 1: Processing chain of the L3-Net embeddings for obtaining decision scores.

2.2. Look, Listen, and Learn Embeddings

The idea of Look, Listen, and Learn (L^3 -Net) embeddings [11, 12] is to detect audio-visual correspondence between a video frame and an audio clip of length 1s. This is done with a neural network consisting of two convolutional networks, an audio subnetwork and a video subnetwork, and a fusion subnetwork that concatenates the vector-sized outputs of the audio and video subnetwork (the embeddings) and predicts whether they belong together or not. By using video frames and audio clips of the same video as positive samples and frames and clips of different videos as negative samples, the entire network can be trained unsupervisedly without requiring any labeled data, which is costly to obtain. After training, audio embeddings are extracted by only using the audio subnetwork. Throughout this paper, the term L^3 -Net embeddings always refers to these audio embeddings. More details can be found in [11, 12].

To extract L^3 -Net embeddings, the open-source implementation openL3 [20] pretrained on the music subset of AudioSet [21] has been used. The embeddings are extracted from log-Mel spectrograms with 256 Mel bins, which in turn are extracted from overlapping windows with a length of 1s and a hop size of 0.1s. For all experiments, an embedding size of 512 has been chosen and all embeddings have been normalized by subtracting the mean and dividing by the standard deviation of the embeddings belonging to the training split of the DCASE 2020 dataset.

2.3. X-vector based system

An x-vector model [22] is the state-of-the-art in speaker recognition. Its purpose is to extract speaker embeddings containing all relevant information about a speaker from audio data. To do this, Mel-frequency cepstral coefficients (MFCCs) [23] are computed and temporal convolutions are applied. The x-vectors are obtained by stacking means and standard deviations of the convolutional output using so-called statistical pooling layers.

In [13], it has been shown that an x-vector model can also be applied to open-set machine listening applications by using L^3 -Net embeddings instead of MFCCs. Therefore, all L^3 -Net embeddings belonging to a single audio file can be combined into a single x-vector that contains all relevant information. The x-vector network is trained for 100 epochs with a batch size of 32 using Adam [17] and is implemented via Keras [18] and Tensorflow [19]. When training the network, only manifold mixup [24] is used to augment the data. Basically, manifold mixup means to apply regular mixup [25] to the data representations of all layers and not just the input layer. To this end, mixup layers with mixing coefficients drawn

Table 1: Architecture of the network for combining embeddings.

Subnetwork	Layer	Output Shape
Preprocessing	Input	(T, 512)
	Mixup	(T, 512)
	Gaussian noise (standard deviation: 0.1)	(T, 512)
X-vector	1D Convolution (kernel size=3, Leaky ReLU: 0.1)	(T, 256)
	Mixup	(T, 256)
	1D Convolution (kernel size=3, Leaky ReLU: 0.1)	(T, 256)
	Mixup	(T, 256)
	1D Convolution (kernel size=5, Leaky ReLU: 0.1)	(T, 256)
	Mixup	(T, 256)
	1D Convolution (kernel size=1, Leaky ReLU: 0.1)	(T, 256)
	Mixup	(T, 256)
	1D Convolution (kernel size=1, Leaky ReLU: 0.1)	(T, 512)
	Mixup	(T, 512)
	Mean	512
	Standard deviation	512
	Concatenation	1024
	Dense (Linear)	256
	Length normalization	256
Classifier	Gaussian noise (standard deviation: 0.1)	256
	Mixup	256
	Leaky ReLU: 0.1	256
	Batch normalization	256
	Dropout (rate: 0.8)	256
	Mixup	256
	Dense (Leaky ReLU: 0.1)	256
	Batch normalization	256
	Dropout (rate: 0.5)	256
	Mixup	256
	Dense (Leaky ReLU: 0.1)	128
	Batch normalization	128
	Mixup	128
	Dense (Softmax)	#Classes

from a uniform distribution have been used. Note that the original manifold mixup technique does only apply mixup at a single randomly chosen layer for each batch. Here, it is always applied at each mixup layer. The x-vector network structure, which strongly resembles the one presented in [13], can be found in Tab. 1.

Since x-vector networks are trained discriminatively and thus intra-class information may be lost, any resulting x-vector is concatenated with the mean of the embeddings this x-vector is derived from. This has been shown to significantly improve the outlier detection performance (see [13]). All x-vectors are further processed with a whitening operation, principal component analysis (PCA) as implemented in [26] and regularized linear discriminant analysis (RLDA) as used in [27] while not reducing the dimension. Applying these transformations in this particular order resulted in the best performance in all conducted experiments, which is the reason why these transformations have been chosen. After length normalization, two-covariance probabilistic linear discriminant analysis (PLDA) [28, 29] as implemented in [30] is used to obtain decision scores. PLDA has the advantage that its output is a log-likelihood ratio comparing the likelihood of two x-vectors belonging to the same class to the likelihood that they do not belong to the same class. This is especially useful when detecting outliers because a fixed threshold can be used to mark machine sounds as anomalous whenever the log-likelihood ratio is below that threshold. It should be emphasized, that the x-vector network as well as the RLDA and PLDA models are trained to discriminate between the exact machine ids instead of the machine types. This led to significantly better performance and is another difference to the baseline system where only a single model is trained for all machine ids belonging to the same machine type. The whole processing chain of the L^3 -Net embeddings is depicted in Figure 1.

Using an x-vector based system instead of autoencoders has many benefits: First and foremost, only one model instead of an additional model for each class needs to be trained. Furthermore, an x-vector based model is trained discriminatively and thus is designed to classify among the classes. Although it is still possible to

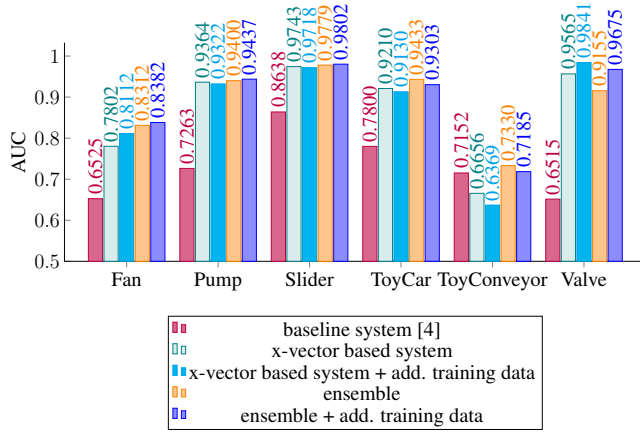


Figure 2: AUCs obtained on the development set with baseline system, x-vector based system and proposed ensemble.

classify with autoencoders by choosing the class that corresponds to the smallest loss, the performance is much worse (see Subsection 3.1). A third benefit is that the input data of the models is much smaller because for autoencoders multiple temporal sections of the spectrogram are stacked and thus the data size is artificially increased. Hence, evaluating the x-vector model is much faster. A possible downside of using an x-vector based model is that one needs to retrain the entire system when adding another class instead of training an additional autoencoder.

2.4. Ensembling strategy

Both models, the baseline model and the x-vector based model, are completely different and are even based on different features. Hence, it seems reasonable that both are making at least some independent errors and thus combining both into an ensemble can increase the performance significantly. An ensembling strategy using logistic regression as used in [31] is not possible because scores obtained with the test split of the development set are not allowed to be used for training. Instead, all relevant information of the sub-systems are concatenated into a single vector before applying PCA, RLDA and PLDA as described in Subsection 2.3. More concretely, this concatenated vector is of the following form:

$$\begin{pmatrix} X := \text{XV}((e_1, \dots, e_K)) \\ \mu_{\text{emb}} := \frac{1}{K} \sum_{k=1}^K e_k \\ \mu_{\text{err}} := \frac{1}{T} \sum_{t=1}^T (\psi_t - \text{AE}(\psi_t))^2 \end{pmatrix} \in \mathbb{R}^{V+S+D(2P+1)} \quad (1)$$

where $(e_k)_{k=1, \dots, K} \subset \mathbb{R}^S$ denote the embeddings belonging to one audio file, $(\psi_t)_{t=1, \dots, T} \subset \mathbb{R}^{D(2P+1)}$ denote all stacked $(2P+1)$ consecutive frames of a log-Mel spectrogram computed from that audio file, XV denotes the x-vector network and AE an autoencoder belonging to the correct machine type. Note that the only difference to the representation used by the x-vector based system is the additional third entry μ_{err} .

3. EXPERIMENTAL RESULTS

3.1. System performances on the development set

The AUCs obtained on the development set are depicted in Fig. 2. One can immediately see that the x-vector based system signif-

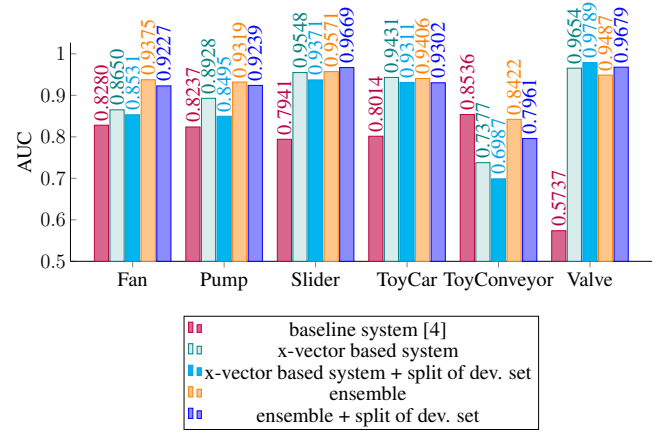


Figure 3: AUCs obtained on the evaluation set with baseline system, x-vector based system and proposed ensemble.

icantly outperforms the baseline system. This is especially true for the machine type “Valve” where the AUC improves from 0.6515 to 0.9565. There is only one machine type, namely “ToyConveyor”, for which the baseline system yields better results than the x-vector based system.

Furthermore, the ensemble performs better than both of its components, even for the machine type “ToyConveyor” where the x-vector based system performed worse than the baseline system. This shows that both models, the baseline system and the x-vector based system, make at least some independent errors. Again, there is only one exception: The AUC for the machine type “Valve” decreases from 0.9565 and 0.9841 to 0.9155 and 0.9675, respectively. This is most likely caused by the relatively poor performance of the baseline system for this particular class.

Another observation to be made is that using additional training data when training the ensemble improves the overall performance only for some machine types while slightly degrading the performance for other machine types (e.g. “ToyConveyor”). This may seem counterintuitive at first, but the machines contained in the additional training dataset do not match those contained in the training and test split of the development set. Note, that the machine types of the machines do match, only the machine ids are different. The systems trained with more data are expected to achieve better performances when encountering sounds of unknown machines of the six machine types. But for the specific machines of the development dataset adding the additional training data can be seen as adding noisy data and thus leads to worse results in some cases.

3.2. System performances on the evaluation set

To determine the performances on the evaluation dataset, all systems have been trained using the additional training set only or in combination with the training split of the development set. The AUCs obtained on the development set are depicted in Fig. 3. They verify all major trends established on the development set: The x-vector based system outperforms the baseline system for all machine types but “ToyConveyor”. For all machine type but “Valve”, the ensemble leads to better results than both of its components. And using additional training data not matching the exact machine ids does not improve the obtained results.

Table 2: Closed-set classification accuracy by machine type.

	baseline system [4]	x-vector based system (PLDA output)	x-vector based system (PLDA output) also trained with add. training data
normal dev. data	93.277%	99.852%	99.741%
anomalous dev. data	76.979%	89.596%	90.949%
all dev. data	85.075%	94.691%	95.317%
evaluation data	81.698%	85.077%	92.707%

Table 3: Closed-set classification accuracy by machine id.

	x-vector based system (PLDA output)	x-vector based system (PLDA output) also trained with add. training data
normal dev. data	99.092%	98.815%
anomalous dev. data	67.819%	64.162%
all dev. data	83.355%	81.377%
evaluation data	machine ids are unknown	74.095%

3.3. Comparison of closed-set classification performance

A good closed-set classification performance is not necessary for detecting anomalous sounds. Still, perfect classification results are useful when operating with the system in practical applications because the user does not need to select which machine (type) is being recorded. This greatly simplifies handling the software or maintenance device. Moreover, it is also possible to obtain the machine type or even exact machine id from the recording without any additional costs. This is especially useful for non-experts who need further information about a machine or experts who need additional information about a specific machine as for example its production year or the date of the last maintenance check.

The closed-set classification accuracies when detecting the machine types can be found in Tab. 2. To evaluate the baseline system, the class corresponding to the autoencoder with the smallest loss has been chosen. As expected, the x-vector based system performs significantly better than the baseline system. The reason is that the x-vector based system is trained discriminatively whereas the baseline system is not. Furthermore, both systems have a higher classification accuracy with normal machine sounds than with anomalous sounds. More concretely, the x-vector based-system has a nearly perfect accuracy for normal sounds and an accuracy of about 90% for anomalous sounds. Here, the reason is that recordings from fully functioning machines sound alike whereas different mechanical failures can alter the sounds in many different ways making it more difficult to recognize the correct machine type. Including training data of additional machines slightly improves the performance in case the machines are not present in the test set and significantly improves the performance when they are present. While the overall results look promising, one needs to keep in mind that there are only six different machine types present in these experiments. In realistic applications, more machine types and thus lower classification accuracies are to be expected.

The closed-set accuracies for detecting the exact machine ids are depicted in Tab. 3. When comparing the results to the ones obtained with the machine types, it is immediately visible that the performance is worse because the task is inherently more difficult. Again, sounds belonging to normal machines are still recognized close to perfectly whereas the performance degrades even more when anomalous sounds are encountered for the same reason as stated above. Here, using additional training data not belonging to the machines present in the dataset slightly degrades the performance. This seems reasonable since knowing additional machines does not help to distinguish the machines one is interested in.

4. CONCLUSIONS AND FUTURE WORK

In this paper, an x-vector based system using L^3 -Net embeddings for anomalous sound detection, adapted from [13], has been presented and evaluated in task 2 of the DCASE challenge 2020. It has been shown that the system significantly outperforms the baseline system when detecting anomalous sounds as well as when detecting the machine type or exact machine id a sound belongs to. Furthermore, an ensemble of the x-vector based system and the baseline system has been presented, which performs even better than both of its components.

In the future, it is planned to try other loss functions for the x-vector network that do not enforce a discriminative behavior on the x-vectors. When detecting anomalous data, a discriminative structure is not needed and might even mask valuable information leading to worse performance [32]. Thus, another loss function and replacing RLDA with a non-discriminative technique as within-class covariance normalization (WCCN) [33, 34] may lead to better performance. Further improvements in terms of performance can probably be gained by replacing the baseline system with a more sophisticated autoencoder architecture.

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